

Will Buziak

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Background

Computer & Mechanical Engineer - Computer Science PhD student

Education

Colorado School of Mines

Doctor of Philosophy

Field of Study: Computer Science

Expected Graduation: Spring 2028

University of Tennessee, Knoxville

Bachelor of Science

Major: Mechanical Engineering

Minor: Computer Science

Graduated: May 2024

Research Interests

Microarchitecture Design, Quantum Computation & Hardware, Multi-Physics Modeling, Heterogeneous Computing, Edge Computing, High Performance Computing

Experience

Bahar Lab

Graduate Research Assistant — Aug 2024 - Present

- Hybrid Quantum-Classical Algorithm development
- Secure Memory Computer Architecture Design for RISC-V Trusted Execution Environments
- Simulation Verification and Modification using the gem5 simulator

References: Dr. Iris Bahar (Mines), Dr. Tamara Silbergleit Lehman (UCB), Dr. Pooja Rao (NVIDIA)

Los Alamos National Lab

Graduate Student Intern — June 2026 - August 2026

- Security research on LLMs and the implications of sponge attacks on hardware

References: Dr. Michael Teti (LANL)

Neuromorphic Computing TENNLab

Undergraduate Research Assistant — Dec 2022 - Aug 2024

- Mathematical modeling on a joint project with ORNL Research Scientists utilizing a neuromorphic controller on a dual fuel diesel-ammonia combustion engine

- Robotic design for swarming robotic applications emphasizing onboard low-power electronics
- Event-based camera data processing for object recognition utilizing speed filtering and clustering algorithms within the TENNLab framework

References: Dr. James Plank (UTK), Dr. Catherine Schuman (UTK), Dr. Charles Rizzo (UTK), Dr. Bryan Maldonado (ORNL), Dr. Brian Kaul (ORNL)

Electrochemical Energy Conversion and Storage Lab

Undergraduate Research Assistant — Aug 2022 - May 2024

- Multi-physics modeling for hydrogen electrolyzer research applications
- Designed a web-based interactive user interface for the visualization of Electric Vertical Take-Off and Landing (eVTOL) vehicle energy storage and power delivery system requirements

References: Dr. Matthew Mench (UTK), Dr. Douglas Aaron (UTK), Dr. Anirban Roy (NREL)

Eck-Lectric Industries

Mechanical Engineering Intern — Oct 2022 - Dec 2022

- Assisted in original product design for patent development

References: Fred Martin, Paul Eck

Shaw Industries

Process Engineering Co-op — May 2022 – Aug 2022

- Led process improvement projects in a manufacturing environment with a focus on waste optimization, automation and safety

References: Ashley Muench

IEEE / ACM Invited Talks

- YARCH '25 Workshop hosted by ASPLOS-EUROSYS 2025
- IEEE Denver Computer, Information Theory, and Robotics (CIR) - Spring 2025

Awards / Scholarships

- C-MAPP Scholar 2025-2026
- EnergyTech University Prize 2023 Bonus Prize finalist

Certificates

- IBM Qiskit Global Summer School 2025 - Quantum Excellence

Computer Skills

Languages: C/C++, Rust, Python, Java, HTML/CSS, MATLAB, RISC-V

Design Software: VHDL, Verilog / SystemVerilog, Solidworks, Onshape

Operating Systems: Linux/Unix, Windows Suite

Version Control: git (Github / Bitbucket / Gitlab)

Single Board Computers: Raspberry Pi 4/5, Raspberry Pi Pico, Arduino

Architecture Simulators: Gem5, QEMU

Software Design Kits: Qiskit, D-Wave, CUDA-Q

Technical Skills

Software: RTL Design, Mathematical & Multi-physics Modeling, Simulators and Emulation Frameworks

Quantum Computing: Quantum Programming, Architecture & Theoretical Quantum Computation

Fabrication: Woodworking, Metal-Working, Additive Manufacturing, Soldering

Textiles: Crocheting

Contact

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- Personal Website: <https://wbuziak.github.io/wbuziak/>